

CUSTOMER MILESTONE BRIEF

MS #47 - Graph-Bound Live Evidence Control Tower

Purpose, customer value, delivered capability, evidence, and claim boundaries

Status	Complete
Release	0.47.0
Date	2026-08-31
Audience	Customers, partners, executives, architects, delivery teams, and assurance reviewers

Executive summary

MS #47 - Graph-Bound Live Evidence Control Tower converts a specific part of the LIGHTYEAR modernization approach into a governed, reviewable capability. Gives customers, operators, and executives a credible live command-post view whose status is visibly bound to the same Knowledge Graph identity used by the underlying evidence.

Why this matters to a customer

Gives customers, operators, and executives a credible live command-post view whose status is visibly bound to the same Knowledge Graph identity used by the underlying evidence.

The practical result is a narrower, evidence-backed decision instead of a broad modernization claim. Commercial, architecture, delivery, security, and assurance teams can review the same capability, proof, and limits.

What the milestone delivers

- A Material-influenced product shell using the investor-deck palette: warm-white work surfaces, a near-black operational plane, gold hierarchy, bronze boundaries, modern sans-serif typography, sentence case, rounded controls, and more space.
- A graph-bound Company to Business problem to Workload to Technology scope to Operator lens path in which the first selections constrain available workloads, the workload changes the graph root, and scope and lens refine the resulting graph.
- A dedicated Knowledge Graph binding card showing canonical content identity, entity and relationship counts, freshness, and downstream binding status.
- A first-class graph operational source covering the canonical graph snapshot and its source-evidence pack.

- Hash-chained graph projection change events on the existing replayable Server-Sent Events plane plus a five-second freshness refresh between events.
- Live graph reload that refreshes metadata, perspectives, operator context, legend, metrics, and the active graph view.
- Fail-closed invalidation of the source-evidence pack, Runtime projection, or Audit projection when its graph identity no longer matches the canonical graph.
- A critical alert for graph-binding invalidation and regression coverage across policy, HTTP, UI, identity, freshness, and documentation contracts.

How it differs from earlier work

MS #17 introduced the live read-only evidence plane, MS #46 added unified-estate navigation and trace, and MS #47 makes the canonical graph itself observable while bringing the interface to the LIGHTYEAR investor visual standard.

Earlier evidence is not automatically promoted into this milestone. A parser, simulator, local comparison, or generated artifact is not live platform equivalence or production readiness.

What a customer can do with it

A customer can use graph-bound live evidence control tower as a bounded decision and delivery capability, attached to approved sources, owners, policies, target architecture, acceptance criteria, and authorized evidence.

Unresolved semantic loss, policy decisions, unsupported behavior, and live-evidence requirements remain visible.

Evidence and acceptance posture

The repository capability is complete because its committed implementation and release record are present. Customer qualification remains claim-specific and evidence-class-specific.

Review exact source and artifact identities, distinguish development from authorized live evidence, inspect policy and loss classifications, confirm passed and blocked gates, and verify that later changes have not invalidated the evidence.

Boundaries and claims not made

- The Control Tower remains a read-only projection and does not gain approve, dispatch, retry, promote, or exception-authoring authority.
- The local reference remains loopback-only and does not provide enterprise SSO, RBAC, managed streaming, or production hosting.
- Graph freshness proves observation of the committed snapshot and evidence pack; it does not prove customer runtime behavior.
- A graph change invalidates mismatched downstream evidence; it does not automatically regenerate or approve that evidence.

- Customer execution evidence, native platform equivalence, and production readiness remain false.

Source of record

- CHANGELOG.md release section(s): 0.47.0
- README.md product and operating guidance
- LIGHTYEAR-ROADMAP.md governing sequence and claim classifications
- docs/milestones/catalog.json metadata and docs/milestones/manifest.json artifact integrity

The underlying schemas, receipts, ledgers, tests, and policy decisions remain authoritative when greater detail is required.